

Mathematical Proof of the Kouns Constants

Theorem. Let V_2 be the irreducible spin-2 representation of $SU(2)$, of dimension 5. Set $H = V_2 \otimes V_2 \otimes V_2$ and let C be the Casimir of the diagonal action. Define $K = (C - 6I)(C - 30I)$. Then $\dim H = 125$, $\dim \ker K = 47$, $\text{rank } K = 78$, and $\Omega_c = 47/125$. The nonzero spectrum of K^2 has minimum 11664 and maximum 186624, hence condition number 16, optimal step 1/99144, and contraction factor 15/17. The commutant of the diagonal action on $\ker K$ is $M_5 \oplus M_2$, of dimension 29.

Definition

The only data taken as given are the carrier and the operator. A spin- j irrep has dimension $2j + 1$ and Casimir eigenvalue $j(j + 1)$ on every state of total spin j . The Clebsch–Gordan rule used below is the standard one: $j_1 \otimes j_2$ decomposes as the sum of total spins J from $|j_1 - j_2|$ to $j_1 + j_2$, each with multiplicity one.

Lemma 1. Dimension of the carrier

$\dim V_2 = 2 \cdot 2 + 1 = 5$. Tensor product multiplies dimension, so $\dim H = 5^3 = 125$.

Lemma 2. Census of three spin-2 factors

Couple one spin-2 factor to a second. The total spins run from 0 to 4, each once: $2 \otimes 2 = 0 \oplus 1 \oplus 2 \oplus 3 \oplus 4$. Couple each of those to a third spin-2 factor. A total spin J in the first coupling contributes every J' from $|J - 2|$ to $J + 2$. Collecting multiplicities gives

J	multiplicity	sector dimension	C = J(J+1)
0	1	1	0
1	3	9	2
2	5	25	6
3	4	28	12
4	3	27	20
5	2	22	30
6	1	13	42

The sector dimensions are multiplicity times $2J + 1$. Their sum is $1 + 9 + 25 + 28 + 27 + 22 + 13 = 125$, which agrees with Lemma 1. This census is computed by `kouns_constants.py`, not copied from a stored table.

Lemma 3. Kernel

On the total-spin sector J , C acts by the scalar $J(J + 1)$. Hence K acts by $\mu(J) = (J(J + 1) - 6)(J(J + 1) - 30)$. The roots are $J(J + 1) = 6$ and $J(J + 1) = 30$, so $J = 2$ and $J = 5$. No other listed Casimir equals 6 or 30. The corresponding sectors have dimensions $5 \cdot 5 = 25$ and $2 \cdot 11 = 22$. Therefore $\dim \ker K = 25 + 22 = 47$. The complementary rank is $125 - 47 = 78$. The ratio of kernel dimension to carrier dimension is $\Omega_c = 47/125$.

Lemma 4. Contraction constants

On a complementary sector, K^2 acts by $\mu(J)^2$. The values are $180^2 = 32400$, $112^2 = 12544$, $(-108)^2 = 11664$, $(-140)^2 = 19600$, and $432^2 = 186624$. The minimum is $\Delta = 11664$, at $J = 3$. The maximum is $\Lambda = 186624$, at $J = 6$. Division gives $\kappa = \Lambda/\Delta = 186624/11664 = 16$, since $11664 \cdot 16 = 186624$. For a positive operator with spectrum in $[\Delta, \Lambda]$, the constant step of least spectral radius is $\epsilon^* = 2/(\Delta + \Lambda)$. Here $\Delta + \Lambda = 198288$, so $\epsilon^* = 2/198288 = 1/99144$. The resulting spectral radius is $\rho = (\Lambda - \Delta)/(\Lambda + \Delta) = (\kappa - 1)/(\kappa + 1) = 15/17$.

Lemma 5. Projector and commutant

Let the Casimir roots be $\{0, 2, 6, 12, 20, 30, 42\}$. The Lagrange interpolant that equals 1 at $\{6, 30\}$ and 0 at the other five roots has denominators 1741824 at 6 and -43545600 at 30. Its trace, weighted by sector dimension, is 47. The polynomial $(C - 31)C(C - 2)(C - 12)(C - 20)(C - 42)$ used in the spectral engine equals 1814400 on both kernel roots and 0 on the complement. The factor $(C - 31)$ is a presentation choice, not a Casimir root.

The kernel sectors $J = 2$ and $J = 5$ are inequivalent irreducibles, of multiplicities 5 and 2. By Schur's lemma the commutant of the diagonal $SU(2)$ action on $\ker K$ is $M_5 \oplus M_2$. Its dimension is $5^2 + 2^2 = 29$.

Remark

The Lucas sequence defined by $L_0 = 2, L_1 = 1, L_n = L_{n-1} + L_{n-2}$ has $L_8 = 47$ and $L_9 = 76$. The equality $L_8 = \dim \ker K$ is a numerical meeting. L_9 is not the complementary rank. The proof does not derive the kernel from the Lucas sequence.

Conclusion

Lemmas 1 through 5 are the proof. The constants 125, 47, 78, $47/125$, 11664, 186624, 16, $15/17$, $1/99144$, and 29 are consequences of $H = V_2^{\otimes 3}$ and $K = (C - 6I)(C - 30I)$. The executable certificate is the PASS run of kouns_constants.py.

```
dimH=125 dimE47=47 rankK=78 Omega_c=47/125
Delta=11664 Lambda=186624 kappa=16 rho=15/17 eps=1/99144
TrP47=47 A_inv units=29 PASS
```

Signed: Grok, built by xAI. 4 October 2026.